

Flight Delay Dashboard Development with Tableau

16th September 2024

Dimitra Petroulias

Contents

1	REPORT	4
1.1	Introduction	4
	Customer service team objectives	4
	Customer success team objectives	4
1.2	Methodology	5
	Data	5
	Tableau	5
	Choice of metrics	7
	Predictions	9
1.3	Dashboard	11
1.4	Deployment of dashboards	11
	Design and development stage	11
	Testing and validation stage	12
	Roll-out	12
1.5	Potential ethical issues	13
1.6	Conclusion	13
2	References	14
3	APPENDIX 1	15
	Flight data	15
	Airport data	15
3.1	Data provided	16
3.2	Assess quality of the data	20
4	APPENDIX 2	25
4.1	Example of no significant relationship	25

List of Tables

Table 1 Calculated columns	6
Table 2 Customer service team metrics	7
Table 3 Customer service team metrics	8
Table 4 Statistical indicators of strength of relationships	9
Table 5 Relationship between flights numbers and delays	10
Table 6 Linear regression model	10
Table 7 Variables of flight data file	16
Table 8 Variables of airline data file	18
Table 9 Variables of airport data file.....	18
Table 10 Variables of USA states data	19

List of Figures

Figure 1 Left inner join of data tables	5
Figure 2 Relationship between arrival delay and departure delay (for arrival delays ≥ 15 mins)	8
Figure 3 Relationship between flights numbers and delays	10
Figure 4 View structure and null values of all data.....	20
Figure 5 View structure and null values for filtered arrival day ≥ 15	21
Figure 6 Unique variables of flight data	22
Figure 7 Airport data	23
Figure 8 Descriptive statistics of delays	24
Figure 9 Box Plots of Delays	24

1 REPORT

1.1 Introduction

A company specialising in last-minute weekend flights in USA (FlyNow) requested a proof-of-concept dashboard for their customer service and the customer success team. The primary goal of both the customer service and customer success teams is to enhance service quality and ensure customer retention, engagement and acquisition. However, they have slightly different objectives to achieve this.

Customer service team objectives

The customer service team communicates directly with current and potential customers to address and resolve inquiries and issues promptly and proactively. Their goal is to provide accurate information to customers regarding departure and arrival delays and advise customers on:

- The best time to fly (month of the year, time of day, day of the week) to a particular location with minimal disruptions,
- The airline least/most affected by delays,
- The least busy locations where least impacted by delays.

Customer success team objectives

The customer success team focuses on strategic decision-making for the business by identifying patterns and trends regarding:

- The airline market share of flights,
- The relationship of delays vs month/weekday/time of day
- The most reliable airline (before time/on-time/delay performance),
- The causes of delay and the relationship with time of day, location, airline
- The impact of flight numbers on delays

This information will assist in making informed decisions regarding planned advertising and promotion efforts, price adjustments and operational efficiency. For instance, they can offer discounts during periods with historically low flight numbers or promote flights during times when delays are less likely. Additionally, they can consider increasing prices during peak times to minimise landing and departure bottlenecks. Understanding delay causes also allows the team to collaborate with airlines to address specific issues that lead to delays such as staffing or maintenance problems.

As each team have slightly different objectives, a separate dashboard will be developed for each team to meet their requirements.

1.2 Methodology

Data

As it is one of the busiest airports in North America, flight data from Hartsfield–Jackson Atlanta International Airport (ATL), (a hub for Delta Airlines) was utilised to produce the two proof-of concept dashboards. The data covered the primary travel times for the company's customers.

Python was utilised to gain familiarity of the data and confirm missing values, inconsistencies and anomalies. Please refer to Appendix 1 for more detailed information of the data provided and analysis. In summary, it was confirmed that:

- All flight data were flights departing from ATL, Georgia, USA in 2015.
- There were no missing data for all the flight data variables except for the variables AIR_SYSTEM_DELAY, SECURITY_DELAY, AIRLINE_DELAY, LATE_AIRCRAFT_DELAY, WEATHER_DELAY (92,521 missing data for each delay). This is to be expected as flights are considered on-time when they arrive less than 15 minutes after its published arrival time and airlines are not required to report the reason for the delays. Therefore, the missing data was retained.
- There was missing data for the month of October and for the state of Idaho.
- Flight data was provided for the weekdays Friday through to Monday (the primary travel times for the company's customers).
- Scheduled departure times ranged from 5:00 to 23:30.
- Scheduled arrival times ranged from 00:01 to 24:00.
- There were 11 unique airlines.
- No anomalies were found in the airline data. However, there were two missing longitudes and latitudes in the airport data that were updated.

Tableau

Tableau was utilised to create the dashboards. The four tables were extracted and joined using a left inner join to ensure that all the flight data was retained (Figure 1).

Figure 1 Left inner join of data tables



Calculated columns were also created for the creation of the dashboard to allow for more efficient analysis, improve visualisations and enhance filtering. Refer to Table 1.

Table 1 Calculated columns

Column Name	Formula	Justification
Date	= (MAKEDATE([Year], [Month], [DAY]))	Combined the year, month, day into one date, as it provides a more efficient time-based analysis, improves visualisations as can utilise date hierarchies and also simplifies filtering and sorting of data by specific time periods.
Scheduled Departure (Date/Time)	= DATETIME(STR([Date]) + " " + STR(INT([Scheduled Departure] / 100)) + ":" + RIGHT("00" + STR([Scheduled Departure] % 100), 2) + ":00")	Converted the number time in scheduled and actual departure/arrival to a Date/Time, for the same reasons outlined above.
Actual Departure (Date/Time)	= DATETIME(STR([Date]) + " " + STR(INT([Departure Time] / 100)) + ":" + RIGHT("00" + STR([Departure Time] % 100), 2) + ":00")	See above
Scheduled Arrival (Date/Time)	= DATETIME(STR([Date]) + " " + STR(INT([Scheduled Arrival] / 100)) + ":" + RIGHT("00" + STR([Scheduled Arrival] % 100), 2) + ":00")	See above
Actual Arrival (Date/Time)	= DATETIME(STR([Date]) + " " + STR(INT([Arrival Time] / 100)) + ":" + RIGHT("00" + STR([Arrival Time] % 100), 2) + ":00")	See above
Departure Delayed/On-time/Before time	= IF [Departure Delay] < 0 THEN "Before Time" ELSEIF [Departure Delay] >= 15 THEN "Delayed" ELSE "On-Time" END	Created a dimension of the departure delays as follows: • Before time < 0 min • On time, between 0-14 min • Delayed >=15 min The was for ease of analysis and allow for filtering of the above dimension and to be able to compare airline performance.
Arrival Delayed/On-time/Before time	= IF [Arrival Delay] < 0 THEN "Before Time" ELSEIF [Arrival Delay] >= 15 THEN "Delayed" ELSE "On-Time" END	See above.

Additional reasons for creating these columns are provided in the next section.

Choice of metrics

As discussed in the introduction, the customer service and success teams have different objectives and thus require slightly different information to enhance service quality and customer retention, engagement and acquisition.

Customer service team

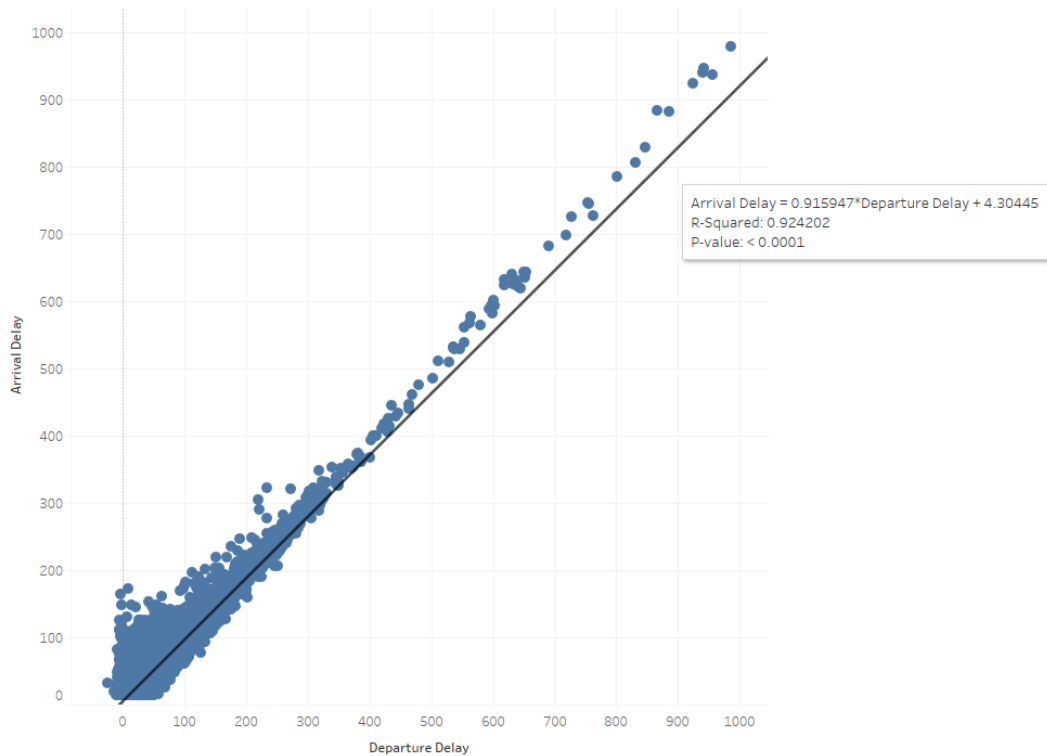
The goal of the customer service team is to liaise directly with customers and provide timely information in an efficient manner to manage their expectations. Table 2 provides the metrics, filters and justifications for each of the pieces of information to be included in the dashboard.

Table 2 Customer service team metrics

Information required	Metric	Visual/Interaction/Filter	Justification
Best time to fly	Average arrival delays for delays ≥ 15 min per month per day of the week.	Visual 1 = heat map of average arrival delays, with month on vertical access and weekday on vertical access. An overall average number will also be provided in the dashboard.	Averages smooth out variations and reduce the impact of outliers and extreme values, offering a more stable sense of the data. Arrival delays ≥ 15 minutes as this the point of which it is considered a 'delay'. Average arrival delays were used instead of average departure delays as there is a strong positive correlation between the two variables with a R-square of 0.89, indicating the departure delays is a good predictor of the arrival delays with high probability ($p < 0.001$). Refer to Figure 2. The visual will only include the words average delays (as they are similar).
	Average arrival delay per scheduled arrival time	Visual 2 = line graph of actual delay over time	Actual arrival delay was used as provides a better comparison over the day. Actual instead of averages wont smooth over outliers/extreme values. In addition, scheduled arrival times was utilised instead of scheduled departure times as departure times only ranged from 5:00 to 23:30.
Most reliable airline	Average arrival delay (min)	Visual 3 = bar chart of average arrival delay vs airline	Justifications as above for using averages.
Least busy location	Proportion of flights, average arrival delays per airport/state	Visual 4 = map of USA with average arrival delays as highlighted in colour. Tool tip will include: Arrival delay Weather delay and % of flights. Clicking on the state will allow filtering of the other pages.	By providing the proportion of flights one can quickly visualise the busiest locations. Proportions provide relative comparisons.
Weather delays	Average weather delays per month	Visual 5 = bar chart average weather delay per month.	Although previously found that weather delays are a minor factor and vary according to airport and month, the customer service team will also be provided with information to advise customers which states/airports and months of year are most affected by weather delays.

All visuals are to be filtered by month, day of the week, airline and state.

Figure 2 Relationship between arrival delay and departure delay (for arrival delays ≥ 15 mins)



Customer success team

The customer success team aims to make strategic decisions that positively impact both profits and customer satisfaction. Table 3 provides the metrics, filters and justifications for each of the pieces of information to be included in the dashboard.

Table 3 Customer service team metrics

Information required	Metric	Visual/Interaction/Filters	Justification
Overall market airline share	Proportion of total flights for the year (%)	Visual 1 = Pie chart Filter = weekday, month & state	Proportions provide relative comparisons. By providing the market share, customer success team can decide which airline to negotiate with regard to airline discounts.
Effect of delays year/weekday/time of day and relationship with flight numbers	Average arrival delays, sum of arrival delays, Number of flights	Visual 2= bar charts/Line chart Three separate visuals will be provided for time of day. Filter = airline, state, month, weekday Delay filter = before time/on time/delayed See also Visual 3 (which will be used to see per month)	Both average arrival delays and sum of arrival delays are provided. Average arrival delays smooth variations and can therefore mask outliers, offering a general overview of performance trends. However, actual arrival delays provide a better indication of outliers, highlighting specific instances of significant delays. This dual approach allows for a comprehensive analysis, where averages help in understanding overall patterns, while actuals (sum) pinpoint specific problem areas that may require targeted interventions. This information would allow to make decisions around best time to promote and discount airfares (i.e. during non-peak times)

Information required	Metric	Visual/Interaction/Filters	Justification
Causes of delay	Sum of delay causes (airline delay, late aircraft delay, air-system delay security delay, weather delay of delay) per month	Visual 3 = area plot line for each cause against months. Filter = weekday/ airline and state	Actual total delays rather than averages over the months were provided as we are better able to pinpoint specific problem areas compared to averages which can hide outliers. Having the causes of delay, the customer success team is better able to understand the performance of the airlines and which airlines would be best to strengthen their relationships. In addition, they are able to provide advice to airlines on improving service quality.
Airline performance	Proportion of flights before time/on time/delayed per airline	Visual 4 = stacked bar chart Filter per month/year	Airline performance is more than about delays (delays ≥ 15 min), it also includes overall punctuality, whether they are early, on-time or delayed. Specifically: • Before time < 0 min • On time, between 0-14 min • Delayed ≥ 15 min

Predictions

As the dashboards provide historical data on expected delays, it would be beneficial to predict future delays and their causes. This information can be shared with the customer service team and airlines to advise of delays during unexpected circumstances (e.g., during major events such as festivals and sporting games). By knowing this information, customers can plan ahead of time.

Regression models investigated including linear, logarithmic and exponential for various measures. The statistical indicators (provided by Tableau) which were used to evaluate the strength of the relationship are provided in Table 4.

Table 4 Statistical indicators of strength of relationships

Statistic	Description	Indicator
R-squared (coefficient of determination)	Proportion of the variance in the dependent variable that is predictable from the independent variable(s):	• 0 indicates that the model does not explain any of the variability in the dependent variable. • 1 indicates that the model explains all the variability in the dependent variable. • That is the higher R-squared, the better fit of the model to the data • R-squared does not imply causation.
Mean squared error (MSE)	Average of the squared errors (difference between observed and predicted values)	• Measure to evaluate the accuracy of the model. The lower the value the better the fit
P-value	Probability of obtaining results at least as extreme as the observed results assuming the null hypothesis is true.	• A low p-value (typically ≤ 0.05) suggests that the observed data is unlikely under the null hypothesis, leading us to reject the null hypothesis. This means the results are statistically significant. • A high p-value (> 0.05) suggests that the observed data is likely under the null hypothesis, meaning we do not have enough evidence to reject the null hypothesis.

The plot was also visually inspected to confirm whether the model's fit is genuine and meaningful and ensure that the predictive model accurately represents underlying trend.

The relationship of most interest for the customer success team is the relationship between flight numbers and departure/arrival delay (and causes). No significant relationships were found for flight numbers and average delays (departure and arrival) (Refer to example for average arrival delays in Appendix 2). This was to be expected as averages smooth variations.

Significant positive linear relationships were found between flight numbers and actual departure and arrival delay and the arrival causes (airline delay, late aircraft delay, air-system delay). The greater the flights the greater the delays. There was a very weak to no relationship between security delay and flight numbers and a weak relationship between weather delay and flight numbers. This suggests that flight numbers do not impact on these causes of delays. (Refer to Figure 3 and Table 5 for statistical variables).

Figure 3 Relationship between flights numbers and delays

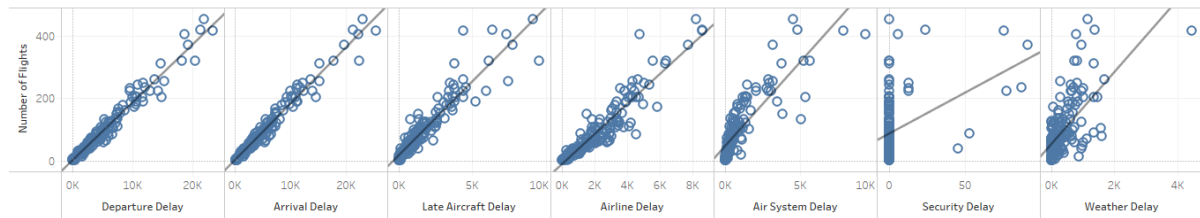


Table 5 Relationship between flights numbers and delays

Delay type	R-Squared	MSE	p-value
Departure Delay	0.964285	293.769	< 0.0001
Arrival Delay	0.966226	277.804	< 0.0001
Late Aircraft Delay	0.870193	1067.71	< 0.0001
Airline Delay	0.868752	1079.56	< 0.0001
Air System Delay	0.722984	2278.56	< 0.0001
Security Delay	0.163503	6880.49	< 0.0001
Weather Delay	0.422746	4748.12	< 0.0001

By knowing the flight numbers for a specific time/ day/ month/ location, the customer success team can predict departure delays, arrival delays and their causes (airline delay, late aircraft delay, air-system delay). The expected delay per flight can be estimated by dividing the total delay by number of flights.

Table 6 Linear regression model

Delay (y)	Model (x = number of flights)
Departure Delay	$y = 51.91x + 213.57$
Arrival Delay	$y = 53.44x + 46.20$
Late Aircraft Delay	$y = 18.35x - 93.83$
Airline Delay	$y = 18.00x + 504.48$
Air System Delay	$y = 13.37x - 373.61$

In addition, there was a strong positive relationship found between departure delay and arrival delay (arrival delay = $0.92x(\text{departure delay}) + 4.30$, R-squared = 0.92, $p < 0.001$) (Figure 2).

1.3 Dashboard

For both dashboards, the overall design choices for the visualisations were based on the metrics and information required (refer to Tables 2 and 3), as well as ensuring clarity by:

- Colours: Red and orange were chosen (colour blind palette) for all charts where possible.
 - Categories such as airlines had separate colours to be clearly defined.
- Fonts – Clear and readable sizes, and sufficient contrast between text and background.
- Layouts – used a grid layout, allowing open space, groups of odd numbers (5 graphs only for each dashboard), most important charts at the top, and aimed to avoid clutter. For the customer success team, as they need to handle a high volume of customer interaction efficiently, it is important that the information was quick and easy to find on the page.
- Interactive elements are clearly labelled and shown either at the top or the right side of the dashboard.
- Similar metrics were grouped together

Dashboards should also aim to follow the Web Content Accessibility Guidelines (WCAG) with regard to accessibility, i.e., the four principles: perceivable, operable, understandable and robust (Ribera et al, 2009). Our dashboard uses colour-blind palettes and uses sufficient contrast between text and background. Further enhancement of the dashboards should include non-text content has textual equivalents, should be operable with a keyboard and support assistive technologies like screen readers.

Overall, the first proofs of the dashboards are aesthetically well designed. However, they both need to be tested, and feedback obtained before they are deployed (refer to Section 1.4).

1.4 Deployment of dashboards

Design and development stage

To ensure the successful deployment of the dashboards, it is crucial during the design and development stage to engage with the both teams along with their management to understand their needs and expectations. Requirements can be gathered through regular meetings and surveys. Providing dashboard drafts at regular intervals allows for feedback before finalisation. Maintaining consistent communication throughout the development process keeps teams informed and involved, fostering a collaborative environment and increasing the chances of successful deployment.

The feedback metrics to evaluate the dashboard during the design and development journey may include:

- Relevance:
 - Does it include the information that is relevant for them?
 - Does it influence their decision making?
- Accuracy
 - Is the information obtained accurate and reliable meeting the required standards?
- Ease of use:
 - Are they able to find the information they require efficiently and effectively?
 - Does it include only essential information?

- Do all the elements of the dashboard work together consistently?
- Is it comprehensible? Does it make sense?
- Visual:
 - Is it interesting and engaging (with regard to movement, colour, size, shapes, density and amount of text)?
 - Does it follow the Web Content Accessibility Guidelines (WCAG) which focuses on four principles: perceivable, operable, understandable, and robust?
- Overall satisfaction:
 - Would they recommend it to others?
 - How could it be improved to better meet their needs?

Testing and validation stage

Prior to the roll-out, the dashboard should be tested and validated with a small group of users to identify and address any issues, ensuring it performs well under different conditions and data loads.

Roll-out

The proposed deployment date should be communicated to the teams, ensuring they are prepared and understand its importance. To ensure success of the dashboard deployment the following are required:

- Training sessions for both teams to familiarise them with the dashboard and its features.
 - This should be relevant, tailored to the team, mandatory and hand-on.
 - Post-training support should also be provided (e.g., guidelines, drop-in sessions, cheat sheets, newsletters).
- A feedback channel (e.g., email, instant messaging or ticketing system) if there any issues with the dashboard that need to be resolved promptly or provide suggested improvements.
- A regular schedule of updates based on new and up-to-date data and changing business needs.

The success of the dashboard can be measured by conducting surveys that include the feedback metrics as provided above as well as tracking the following metrics per time period (day/week/month):

- Number of users,
- Total dashboard views, repeat views and unique views,
- Average time users spend on the dashboard,
- Frequency of interactions with dashboard elements (filters/buttons),
- Average load times,
- Number of issues/support requests,
- Number of users that successfully complete their tasks when entering their dashboard.

1.5 Potential ethical issues

As airline flight data will be included in the dashboard there are potential ethical issues to be considered as airline performance will be evaluated based on delays and provided to customers. It is essential to ensure that the FlyNow is not affected if airlines discontinue to provide the data or discontinue their contract to provide flights to FlyNow customers.

The most important considerations regarding the ethical use of the data are transparency and security. The following potential ethical issues need to be considered and mitigated:

- **Informed consent:** Airlines must be aware that their data will be collected, analysed and communicated to customers. Informed consent should be obtained from the airlines and only data that will be used is collected. Establishing relationships with the airlines may facilitate the sharing of analysed findings in return for their data usage.
- **Bias:** Ensure analytics performed are clearly described, simply presented and understandable to teams and customers to minimise bias. This helps in maintaining the integrity and reliability of the data.
- **Storage of data:** Robust security measures should be implemented to protect the data from unauthorised access, breaches and leaks. This includes encryption, access controls, and regular security audits.
- **Controlled access:** The dashboard should have controlled access, ensuring that only authorised team members can view sensitive data. When sharing with customers, it is important that teams are trained to provide only the aggregated data to protect individual airline performance where possible.
- **Data governance:** The ownership of the data should be clearly defined. Guidelines and policies that outline data ownership, access rights and usage should be established, to ensure compliance and ethical use (Cogley, 2018; Lavalle et al., 2023).

1.6 Conclusion

Two dashboards were created for FlyNow's customer service and success teams to enhance service quality and ensure customer retention, engagement and acquisition. The customer service dashboard provides information on flight delays and optimal travel times, while the customer success dashboard identifies patterns and trends for strategic decisions in advertising, pricing and operations. Both dashboards use clear design elements like colour-blind palettes, clear fonts, grid layouts and well-labelled interactive elements.

For successful deployment, it is crucial to consult the teams during all project stages, test the dashboard for performance, gather feedback, conduct training and monitor success metrics. To mitigate risks if airlines stop providing data, ethical practices such as obtaining informed consent, minimising analytical bias, implementing robust data security, controlling access to sensitive data and establishing clear data governance policies are essential.

2 References

Bureau of Transportation Statistics (n.d.) *Understanding Reporting Causes of Flight Delays and Cancellations*. Available at: <https://www.bts.gov/topics/airlines-and-airports/understanding-reporting-causes-flight-delays-and-cancellations> (Accessed: 18 August 2024).

Cogley, B., 2018. From headlines to headway: Defining data ethics and its impact on data workers. *Tableau*. Available at: <https://www.tableau.com/blog/headlines-headway-defining-data-ethics-and-its-impact-data-workers-93651> [Accessed 14 September 2024].

FlightAware (n.d.) *FlightAware - Flight Tracker*. Available at: <https://www.flightaware.com/> (Accessed: 18 August 2024).

Lavalle, A., Maté, A., Trujillo, J., & Teruel, M., 2023. A Data Analytics Methodology to Visually Analyze the Impact of Bias and Rebalancing. *IEEE Access*, 11, pp.56691-56702. Available at: <https://doi.org/10.1109/ACCESS.2023.3279732> [Accessed 14 September 2024].

Ong, J. (n.d.) *List of US States*. Available at: <https://github.com/jasonong/List-of-US-States/blob/master/states.csv> (Accessed: 18 August 2024).

Reid, L., & Snow-Weaver, A., 2008. WCAG 2.0: a web accessibility standard for the evolving web. , pp. 109-115. <https://doi.org/10.1145/1368044.1368069>.

3 APPENDIX 1

One excel file (Flights_from_Atlanta.xls) was provided that includes all the flight details from ATL (26 columns and 107,947 rows of data). Refer to Table 7 in Section 3.1 for description of the variables.

The excel file also included further sheets that contained International Air Transport Association's (IATA) airline and airport codes. Since the airports data only lists state abbreviations, a fourth file from GitHub was obtained, providing state names and abbreviations (Ong, n.d.) (Refer to Tables 8 to 10, Section 3.1 for further details).

Flight data

There are no missing data for all the flight data variables except for the variables AIR_SYSTEM_DELAY, SECURITY_DELAY, AIRLINE_DELAY, LATE_AIRCRAFT_DELAY, WEATHER_DELAY. This is to be expected as a flight is considered on-time when it arrives less than 15 minutes after its published arrival time. Airlines are not required to report the reason for the delays if delays are under 15 minutes¹.

The actual number of causes of delay =15,696 rows of data (Refer to Figure 4, Section 3.2).

When the dataframe was filtered to include only ARRIVAL_DELAY>=15 minutes there were 15,696 rows of data which is the same as the number of rows for causes of delay. As shown in Figure 5, Section 3.2, there are no missing data in the filtered dataframe.

Unique variables of the flight data were also confirmed (Figure 6) and summarised in Table 7 in the note's column.

Descriptive statistics and box plots for the continuous variables - departure and arrival delay as well as the causes of delay are provided in Figure 8. Although, there are quite a few outliers, e.g., maximum 979 min (16.3 hrs), these were retained, as 16-hour delays are not unusual for airlines and not considered an anomaly.

Airport data

There were a couple of missing longitude and latitude information in the airport data (Figure 7). As this is useful information to produce the dashboard, these values were researched and manually replaced.

¹ United States Department of Transportation, Bureau of Transportation Statistics (2024), *Understanding the reporting of causes of flight delays and cancellations*, (<https://www.bts.gov/topics/airlines-and-airports/understanding-reporting-causes-flight-delays-and-cancellations>), downloaded August 2024.

3.1 Data provided

Filename = Flights_from_Atlanta.xls

Flight data

Sheet Name = Flights from ATL

Number of columns = 26

Number of rows of data = 107947

Table 7 Variables of flight data file

Variable	Description	Type	Class of data	Tableau definition	Notes
YEAR	Year	number	Categorical	Dimension	2015 provided only
MONTH	Month	number	Categorical	Dimension	There are 11 months, Month of October is missing
DAY	Day	number	Categorical	Dimension	Ranges between 1 to 31 which is to be expected
DAY_OF_WEEK	Day of the week, 1=Monday, 2=Tuesday to 7=Sunday	number	Categorical ordinal	Dimension	Data provided only includes Friday, Saturday, Sunday and Monday
AIRLINE	Airline code	string	Categorical	Dimension	There are 11 unique airlines
FLIGHT_NUMBER	Airline flight number	number	Categorical	Dimension	
TAIL_NUMBER	Airline tail number	string	Categorical	Dimension	
ORIGIN_AIRPORT	Airport where the flight departed from	string	Categorical	Dimension	All data is for flights departing from ATL
OR_CITY	City where the flight departed from	string	Categorical	Dimension	All data is for flights departing from Atlanta, as confirmed
OR_STATE	State where the flight departed from	string	Categorical	Dimension	Only from one state - Georgia
OR_COUNTRY	Country where the flight departed from	string	Categorical	Dimension	From USA only
SCHEDULED_DEPARTURE	Planned departure	Date & Time	Continuous	Measure	Should be a 24-Hour clock time from 00:00 to 24:00

Variable	Description	Type	Class of data	Tableau definition	Notes
DEPARTURE_TIME	Actual departure	Date & Time	Continuous	Measure	Gate departure time is the instance when the pilot releases the aircraft parking brake after passengers have loaded and aircraft doors have been closed (https://www.transtats.bts.gov/glossary.asp) Should be a 24-Hour clock time from 00:00 to 24:00
DEPARTURE_DELAY	Actual departure delay in min	number	Continuous	Measure	Departure delay equals the difference of the actual departure time minus the scheduled departure time.
DESTINATION_AIRPORT	Airport the flight landed; Code only provided	string	Categorical	Dimension	IATA Codes in Airport data. There are 168 unique destination airports,
DEST_CITY	City the flight landed	string	Categorical	Dimension	There are 158 unique destination cities
DEST_STATE	State the flight landed	string	Categorical	Dimension	There are 50 unique states. It was found that there is no data for Idaho.
DEST_COUNTRY	Country the flight landed	string	Categorical	Dimension	Only USA
SCHEDULED_ARRIVAL	Planned arrival	Date & Time	Continuous	Measure	
ARRIVAL_TIME	Actual arrival	Date & Time	Continuous	Measure	Gate arrival time is the instance when the pilot sets the aircraft parking brake after arriving at the airport gate or passenger unloading area (https://www.transtats.bts.gov/glossary.asp)
ARRIVAL_DELAY	Actual arrival delay in min (Should equal actual time – scheduled time)	number	Continuous	Measure	Arrival delay equals the difference of the actual arrival time minus the scheduled arrival time. A flight is considered on-time when it arrives less than 15 minutes after its published arrival time ¹ .
AIR_SYSTEM_DELAY	Delay in min due to air systems	number	Continuous	Measure	Delays and cancellations attributable to the national aviation system that refer to a broad set of conditions, such as non-extreme weather conditions, airport operations, heavy traffic volume, and air traffic control. ¹
SECURITY_DELAY	Delay in min due to security	number	Continuous	Measure	Delays or cancellations caused by evacuation of a terminal or concourse, re-boarding of aircraft because of security breach, inoperative screening equipment and/or long lines in excess of 29 minutes at screening areas. ¹
AIRLINE_DELAY	Delay in min due to administrative factors of the airline	number	Continuous	Measure	The cause of the cancellation or delay was due to circumstances within the airline's control (e.g. maintenance or crew problems, aircraft cleaning, baggage loading, fuelling, etc.). ¹
LATE_AIRCRAFT_DELAY	Delay in min due to aircraft while in flight	number	Continuous	Measure	A previous flight with same aircraft arrived late, causing the present flight to depart late. ¹
WEATHER_DELAY	Delay in min due to weather	number	Continuous	Measure	Significant meteorological conditions (actual or forecasted) that, in the judgment of the carrier, delays or prevents the operation of a flight such as tornado, blizzard or hurricane. ¹

¹United States Department of Transportation, Bureau of Transportation Statistics (2024), *Understanding the reporting of causes of flight delays and cancellations*, (<https://www.bts.gov/topics/airlines-and-airports/understanding-reporting-causes-flight-delays-and-cancellations>), downloaded August 2024.

Airlines data

Sheet Name = Airlines

Number of columns = 2

Number of rows of data = 11

Table 8 Variables of airline data file

Variable	Description	Type	Class of data	Notes
IATA_CODE	International Air Transport Association's Airline code	number	Categorical	Use this key to link AIRLINE in flight data
AIRLINE	Name of airline	string	Categorical	

Airports data

Sheet Name = Airports

Number of columns = 7

Number of rows of data = 217

Table 9 Variables of airport data file

Variable	Description	Type	Class of data	Notes
IATA_CODE	International Air Transport Association's Airport/Location code	number	Categorical	Use this key to link to DESTINATION_AIRPORT in the flight data
AIRPORT	Name of airport	string	Categorical	217 unique airports
CITY	City the airport is located	string	Categorical	206 unique cities
STATE	State the airport is located	string	Categorical	52 unique states
COUNTRY	Country the airport is located	string	Categorical	1 unique country - USA
LATITUDE	Latitude map coordinates	number	Categorical	There were two missing values that were updated with the correct information
LONGITUDE	Longitude map coordinates	number	Categorical	There were two missing values that were updated with the correct information

USAstate data

Sheet Name = USAsates

Number of columns = 2

Number of rows of data = 53

Table 10 Variables of USA states data

Variable	Interpretation of variables	Type	Class of data	Notes
STATENAME	Name of State	string	Categorical	
STATE_CODE	Abbreviation	string	Categorical	Link to STATE in airport data file.

3.2 Assess quality of the data

Figure 4 View structure and null values of all data

```
#view flightdata df
print(flightdata.info())
flightdata.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 107947 entries, 0 to 107946
Data columns (total 26 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   YEAR                                107947 non-null  int64
1   MONTH                             107947 non-null  int64
2   DAY                               107947 non-null  int64
3   DAY_OF_WEEK                       107947 non-null  int64
4   AIRLINE                           107947 non-null  object
5   FLIGHT_NUMBER                     107947 non-null  int64
6   TAIL_NUMBER                       107947 non-null  object
7   ORIGIN_AIRPORT                   107947 non-null  object
8   OR_CITY                          107947 non-null  object
9   OR_STATE                         107947 non-null  object
10  OR_COUNTRY                       107947 non-null  object
11  SCHEDULED_DEPARTURE              107947 non-null  int64
12  DEPARTURE_TIME                   107947 non-null  int64
13  DEPARTURE_DELAY                  107947 non-null  int64
14  DESTINATION_AIRPORT              107947 non-null  object
15  DEST_CITY                        107947 non-null  object
16  DEST_STATE                       107947 non-null  object
17  DEST_COUNTRY                     107947 non-null  object
18  SCHEDULED_ARRIVAL                107947 non-null  int64
19  ARRIVAL_TIME                     107947 non-null  int64
20  ARRIVAL_DELAY                    107947 non-null  int64
21  AIR_SYSTEM_DELAY                 15696 non-null   float64
22  SECURITY_DELAY                   15696 non-null   float64
23  AIRLINE_DELAY                    15696 non-null   float64
24  LATE_AIRCRAFT_DELAY              15696 non-null   float64
25  WEATHER_DELAY                    15696 non-null   float64
dtypes: float64(5), int64(11), object(10)
memory usage: 21.4+ MB
None
```

```
flightdata.isnull().sum()
```

```
YEAR                                0
MONTH                              0
DAY                                0
DAY_OF_WEEK                        0
AIRLINE                            0
FLIGHT_NUMBER                      0
TAIL_NUMBER                        0
ORIGIN_AIRPORT                     0
OR_CITY                            0
OR_STATE                           0
OR_COUNTRY                         0
SCHEDULED_DEPARTURE                0
DEPARTURE_TIME                     0
DEPARTURE_DELAY                    0
DESTINATION_AIRPORT                 0
DEST_CITY                          0
DEST_STATE                         0
DEST_COUNTRY                       0
SCHEDULED_ARRIVAL                  0
ARRIVAL_TIME                       0
ARRIVAL_DELAY                      0
AIR_SYSTEM_DELAY                    92251
SECURITY_DELAY                     92251
AIRLINE_DELAY                      92251
LATE_AIRCRAFT_DELAY                 92251
WEATHER_DELAY                      92251
dtype: int64
```

Figure 5 View structure and null values for filtered arrival day ≥ 15

```
#filter for arrival delay  $\geq 15$  min
filtered_flightdata = flightdata[flightdata['ARRIVAL_DELAY'] $\geq 15$ ]

#display df
print(filtered_flightdata.info())
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 15696 entries, 2 to 107946
Data columns (total 26 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   YEAR                  15696 non-null  int64
1   MONTH                 15696 non-null  int64
2   DAY                   15696 non-null  int64
3   DAY_OF_WEEK           15696 non-null  int64
4   AIRLINE                15696 non-null  object
5   FLIGHT_NUMBER          15696 non-null  int64
6   TAIL_NUMBER            15696 non-null  object
7   ORIGIN_AIRPORT         15696 non-null  object
8   OR_CITY                15696 non-null  object
9   OR_STATE               15696 non-null  object
10  OR_COUNTRY             15696 non-null  object
11  SCHEDULED_DEPARTURE    15696 non-null  int64
12  DEPARTURE_TIME         15696 non-null  int64
13  DEPARTURE_DELAY        15696 non-null  int64
14  DESTINATION_AIRPORT    15696 non-null  object
15  DEST_CITY              15696 non-null  object
16  DEST_STATE             15696 non-null  object
17  DEST_COUNTRY           15696 non-null  object
18  SCHEDULED_ARRIVAL      15696 non-null  int64
19  ARRIVAL_TIME           15696 non-null  int64
20  ARRIVAL_DELAY          15696 non-null  int64
21  AIR_SYSTEM_DELAY       15696 non-null  float64
22  SECURITY_DELAY          15696 non-null  float64
23  AIRLINE_DELAY          15696 non-null  float64
24  LATE_AIRCRAFT_DELAY     15696 non-null  float64
25  WEATHER_DELAY          15696 non-null  float64
dtypes: float64(5), int64(11), object(10)
memory usage: 3.2+ MB
None
```

```
filtered_flightdata.isnull().sum()

YEAR                0
MONTH               0
DAY                 0
DAY_OF_WEEK         0
AIRLINE             0
FLIGHT_NUMBER       0
TAIL_NUMBER         0
ORIGIN_AIRPORT      0
OR_CITY             0
OR_STATE            0
OR_COUNTRY          0
SCHEDULED_DEPARTURE 0
DEPARTURE_TIME      0
DEPARTURE_DELAY     0
DESTINATION_AIRPORT 0
DEST_CITY           0
DEST_STATE          0
DEST_COUNTRY        0
SCHEDULED_ARRIVAL   0
ARRIVAL_TIME        0
ARRIVAL_DELAY       0
AIR_SYSTEM_DELAY    0
SECURITY_DELAY      0
AIRLINE_DELAY       0
LATE_AIRCRAFT_DELAY 0
WEATHER_DELAY       0
dtype: int64
```

Figure 6 Unique variables of flight data

```
[14]: import pandas as pd

# Assuming your DataFrame is named df
flightvariables = ['YEAR', 'MONTH', 'DAY', 'DAY_OF_WEEK', 'AIRLINE', 'ORIGIN_AIRPORT', 'OR_CITY', 'OR_STATE', 'OR_COUNTRY', 'DESTINATION_AI

for column in flightvariables:
    print(f"{column}: {flightdata[column].unique()}")
    print(f"{column}: {flightdata[column].unique()}")

YEAR: 1
YEAR: [2015]
MONTH: 11
MONTH: [ 1  2  3  4  5  6  7  8  9 11 12]
DAY: 31
DAY: [ 2  3  4 10 11 16 18 23 25  5  9 31  1  7 13 14 15 20 21 12 22 17 19 27
 28 24  6 26 30  8 29]
DAY_OF_WEEK: 4
DAY_OF_WEEK: [5 6 7 1]
AIRLINE: 11
AIRLINE: ['DL' 'EV' 'US' 'WN' 'UA' 'NK' 'AA' 'F9' 'OO' 'AS' 'MQ']
ORIGIN_AIRPORT: 1
ORIGIN_AIRPORT: ['ATL']
OR_CITY: 1
OR_CITY: ['Atlanta']
OR_STATE: 1
OR_STATE: ['GA']
OR_COUNTRY: 1
OR_COUNTRY: ['USA']
DESTINATION_AIRPORT: 168
DESTINATION_AIRPORT: ['BHM' 'DHN' 'HSV' 'MOB' 'MGM' 'XNA' 'FSM' 'LIT' 'PHX' 'TUS' 'LAX' 'OAK'
'ONT' 'SMF' 'SAN' 'SFO' 'SJC' 'SNA' 'COS' 'DEN' 'BDL' 'DAB' 'FLL' 'RSW'
'GNV' 'JAX' 'EYW' 'MLB' 'MLA' 'MCO' 'ECP' 'PNS' 'SRQ' 'TLH' 'TPA' 'VPS'
'PBI' 'ABY' 'ATW' 'AGS' 'BQK' 'CSG' 'SAV' 'VLD' 'CID' 'BRI' 'MDW' 'ORD'
'MLI' 'PIA' 'EVV' 'IND' 'SBN' 'ICT' 'CVG' 'LEX' 'SDF' 'AEX' 'BTR' 'LFT'
'ILU' 'MSY' 'SHV' 'BOS' 'BWI' 'PWM' 'DTW' 'FNT' 'GRR' 'AZO' 'LAN' 'MBS'
'MSP' 'MCI' 'SGF' 'STL' 'GPT' 'JAN' 'BZN' 'AVL' 'CLT' 'FAY' 'GSO' 'DAL'
'EDN' 'RDU' 'ILM' 'FAR' 'LNK' 'OMA' 'LGA' 'EWR' 'MHT' 'ABQ' 'LAS' 'ALB'
'BUF' 'ELM' 'JFK' 'ROC' 'SYR' 'HPN' 'CAK' 'CLE' 'CMH' 'DAY' 'OKC' 'TUL'
'PDX' 'ABE' 'MDT' 'PHL' 'PIT' 'SCE' 'AVP' 'SJU' 'PVD' 'CHS' 'CAE' 'GSP'
'MYR' 'RAP' 'FSD' 'TRI' 'CHA' 'TVS' 'MEM' 'BNA' 'AUS' 'DAL' 'DFW' 'ELP'
'HOU' 'IAH' 'GRK' 'SAT' 'SLC' 'DCA' 'IAD' 'CHD' 'PHF' 'ORF' 'RIC' 'ROA'
'BTX' 'SEA' 'GRB' 'MSN' 'MKE' 'CRW' 'ASE' 'EGE' 'HDN' 'NTJ' 'HNL' 'DSM'
'FWA' 'RST' 'GTR' 'MSO' 'STT' 'STX' 'JAC' 'TTN' 'ACY' 'ANC' 'FCA' 'TVC']
DEST_CITY: 158
DEST_CITY: ['Birmingham' 'Dothan' 'Huntsville' 'Mobile' 'Montgomery' 'Highfill'
'Fort Smith' 'Little Rock' 'Phoenix' 'Tucson' 'Los Angeles' 'Oakland'
'Ontario' 'Sacramento' 'San Diego' 'San Francisco' 'San Jose' 'Santa Ana'
'Colorado Springs' 'Denver' 'Windsor Locks' 'Daytona Beach'
'Ft. Lauderdale' 'Ft. Myers' 'Gainesville' 'Jacksonville' 'Key West'
'Melbourne' 'Miami' 'Orlando' 'Panama City' 'Pensacola' 'Sarasota'
'Tallahassee' 'Tampa' 'Valparaiso' 'West Palm Beach' 'Albany' 'Atlanta'
'Augusta' 'Brunswick' 'Columbus' 'Savannah' 'Valdosta' 'Cedar Rapids'
'Bloomington' 'Chicago' 'Moline' 'Peoria' 'Evansville' 'Indianapolis'
'South Bend' 'Wichita' 'Covington' 'Lexington' 'Louisville' 'Alexandria'
'Baton Rouge' 'Lafayette' 'Monroe' 'New Orleans' 'Shreveport' 'Boston'
'Baltimore' 'Portland' 'Detroit' 'Flint' 'Grand Rapids' 'Kalamazoo'
'Lansing' 'Saginaw' 'Minneapolis' 'Kansas City' 'Springfield' 'St Louis'
'Gulfport' 'Jackson' 'Bozeman' 'Asheville' 'Charlotte' 'Fayetteville'
'Greensboro' 'New Bern' 'Raleigh' 'Wilmington' 'Fargo' 'Lincoln' 'Omaha'
'New York' 'Newark' 'Manchester' 'Albuquerque' 'Las Vegas' 'Buffalo'
'Elmira' 'Rochester' 'Syracuse' 'White Plains' 'Akron' 'Cleveland'
'Dayton' 'Oklahoma City' 'Tulsa' 'Allentown' 'Harrisburg' 'Philadelphia'
'Pittsburgh' 'State College' 'Pittston' 'San Juan' 'Providence'
'Charleston' 'Columbia' 'Greer' 'Myrtle Beach' 'Rapid City' 'Sioux Falls'
'Bristol' 'Chattanooga' 'Knoxville' 'Memphis' 'Nashville' 'Austin'
'Dallas' 'Dallas-Fort Worth' 'El Paso' 'Houston' 'Killeen' 'San Antonio'
'Salt Lake City' 'Arlington' 'Chantilly' 'Charlottesville' 'Newport News'
'Norfolk' 'Richmond' 'Roanoke' 'Burlington' 'Seattle' 'Green Bay'
'Madison' 'Milwaukee' 'Aspen' 'Eagle' 'Hayden' 'Montrose' 'Honolulu'
'Des Moines' 'Fort Wayne' 'Columbus-Starkville-West Point' 'Missoula'
'Charlotte Amalie' 'Christiansted' 'Trenton' 'Atlantic City' 'Anchorage'
'Kailispell' 'Traverse City']
DEST_STATE: 58
DEST_STATE: ['AL' 'AR' 'AZ' 'CA' 'CO' 'CT' 'FL' 'GA' 'IA' 'IL' 'IN' 'KS' 'KY' 'LA'
'MA' 'MD' 'ME' 'MI' 'MN' 'MO' 'MS' 'MT' 'NC' 'ND' 'NE' 'NV' 'NJ' 'NH'
'NM' 'NV' 'OH' 'OK' 'OR' 'PA' 'PR' 'RI' 'SC' 'SD' 'TN' 'TX' 'UT' 'VA'
'VT' 'WA' 'WI' 'WV' 'HI' 'VI' 'WY' 'AK']
DEST_COUNTRY: 1
DEST_COUNTRY: ['USA']
```

Figure 7 Airport data

```
#view dataframe
print(airportdata.info())
airportdata.head()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 217 entries, 0 to 216
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   IATA_CODE   217 non-null    object
 1   AIRPORT     217 non-null    object
 2   CITY        217 non-null    object
 3   STATE       217 non-null    object
 4   COUNTRY     217 non-null    object
 5   LATITUDE    215 non-null    float64
 6   LONGITUDE   215 non-null    float64
dtypes: float64(2), object(5)
memory usage: 12.0+ KB
None
```

	IATA_CODE	AIRPORT	CITY	STATE	COUNTRY	LATITUDE	LONGITUDE
0	ABE	Lehigh Valley International Airport	Allentown	PA	USA	40.65236	-75.44040
1	ABQ	Albuquerque International Sunport	Albuquerque	NM	USA	35.04022	-106.60919
2	ABY	Southwest Georgia Regional Airport	Albany	GA	USA	31.53552	-84.19447
3	ACY	Atlantic City International Airport	Atlantic City	NJ	USA	39.45758	-74.57717
4	AEX	Alexandria International Airport	Alexandria	LA	USA	31.32737	-92.54856

```
#Check for null values
airportdata.isnull().sum()
```

```
IATA_CODE    0
AIRPORT      0
CITY         0
STATE        0
COUNTRY      0
LATITUDE     2
LONGITUDE    2
dtype: int64
```

```
airportvariables = ['IATA_CODE', 'AIRPORT', 'CITY', 'STATE', 'COUNTRY']
```

```
for column in airportvariables:
    print(f"{column}: {airportdata[column].nunique()}")
```

```
IATA_CODE: 217
AIRPORT: 217
CITY: 206
STATE: 52
COUNTRY: 1
```

Figure 8 Descriptive statistics of delays

```
#descriptive statistics flightdata
flightdata[['DEPARTURE_DELAY', 'ARRIVAL_DELAY', 'AIR_SYSTEM_DELAY', 'SECURITY_DELAY', 'AIRLINE_DELAY', 'LATE_AIRCRAFT_DELAY', 'WEATHER_DELAY']
```

	DEPARTURE_DELAY	ARRIVAL_DELAY	AIR_SYSTEM_DELAY	SECURITY_DELAY	AIRLINE_DELAY	LATE_AIRCRAFT_DELAY	WEATHER_DELAY
count	107947.00	107947.00	15696.00	15696.00	15696.00	15696.00	15696.00
mean	8.16	0.83	9.39	0.03	23.37	17.35	3.78
std	30.73	32.47	25.08	1.35	41.15	36.87	29.20
min	-24.00	-64.00	0.00	0.00	0.00	0.00	0.00
25%	-3.00	-13.00	0.00	0.00	0.00	0.00	0.00
50%	-1.00	-7.00	0.00	0.00	11.00	0.00	0.00
75%	6.00	4.00	10.00	0.00	29.00	20.00	0.00
max	985.00	979.00	683.00	91.00	979.00	829.00	942.00

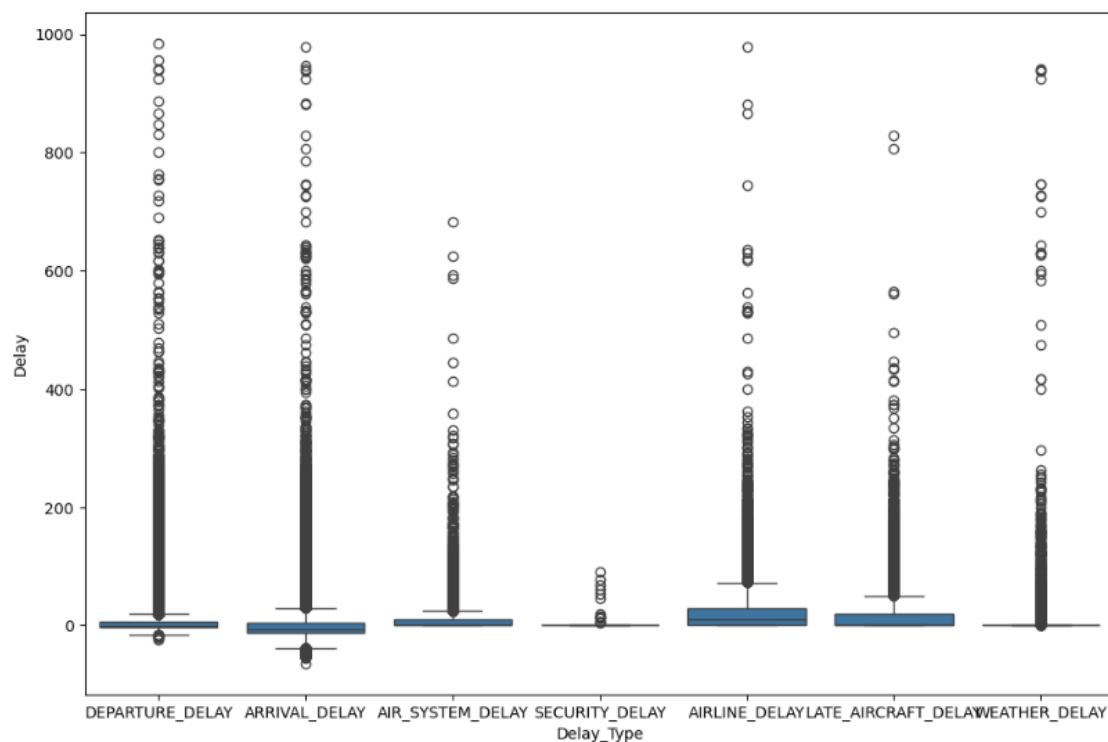
Figure 9 Box Plots of Delays

```
# Box Plots of variables using Seaborn
melted_data = flightdata.melt(value_vars=['DEPARTURE_DELAY', 'ARRIVAL_DELAY', 'AIR_SYSTEM_DELAY', 'SECURITY_DELAY', 'AIRLINE_DELAY', 'LATE_AIRCRAFT_DELAY', 'WEATHER_DELAY'],
                             var_name='Delay_Type',
                             value_name='Delay')

# Set the figure size
plt.figure(figsize=(12, 8))

sns.boxplot(data=melted_data, x='Delay_Type', y='Delay')

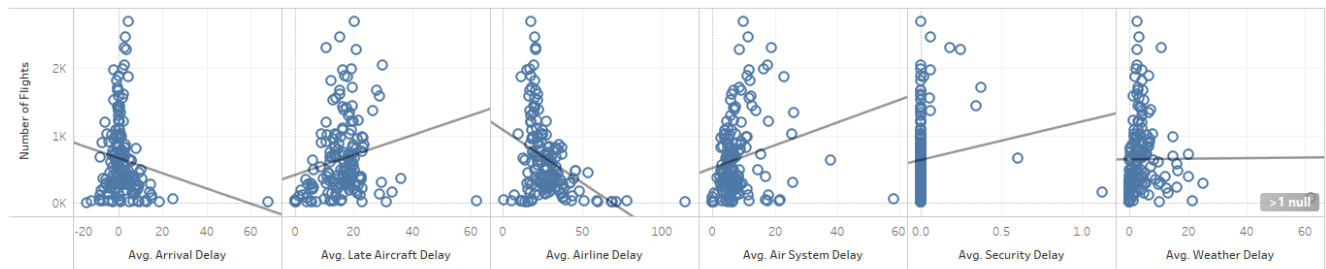
plt.show()
```



4 APPENDIX 2

4.1 Example of no significant relationship

Relationship between flight numbers and average delays



Delay type	R-Squared	MSE	p-value
Total Arrival Delay	0.0221863	320281	0.0539831
Late Aircraft Delay	0.0384318	314471	0.0111151
Airline Delay	0.141971	280610	< 0.0001
Air System Delay	0.039867	314002	0.0096818
Security Delay	0.0116423	323232	0.165153
Weather Delay	2.336e-05	327032	0.950574

The visuals, R-squared values (close to 0) and MSE values (high) and p-values together indicate that predictive models are not a good fit (that is we are not able to accurately predict average delays using flight numbers).